

Evaluating a Parameter-Free Quantum Fidelity Kernel for Text Security Under Representation and Domain Shift

Abstract

Quantum kernel methods map classical text into exponentially large Hilbert spaces, motivated by theoretical conjectures that quantum state fidelity may provide non-linear decision boundaries or enhanced generalization under distribution shift. We ask a narrower question: under controlled matched conditions, does an unparameterized quantum fidelity kernel (two-layer cyclic ZZFeatureMap) provide a consistent practical advantage over matched classical Gaussian RBF kernels for text security? We study this across three curated cybersecurity corpora comprising 153,410 usable text records (SMS Spam Collection, CEAS 2008 Email Corpus, and the multi-source MeAJOR archive) using frozen, leakage-safe protocols across 10 computational seeds. Under matched in-distribution (IID) scaling (2–12 qubits), the quantum kernel is competitive with classical RBF, displaying minor statistically detectable differences at intermediate dimensions (+0.46 percentage points at 8D, $p = 0.0016$; +0.57 pp at 10D, $p = 0.0052$) that remain strictly within the predefined practical-equivalence threshold ($\epsilon = 0.01$ F1) under Two One-Sided Tests (TOST), converging to complete parity at 12D (+0.14 pp, $p = 0.2824$). Against a validation-tuned RBF baseline, the quantum margin narrows to full statistical parity across all dimensions. Under cross-source domain transfer (TREC 2007 $\rightarrow$ TREC 2005/2006), the quantum kernel exhibits a statistically significant performance deficit ($\Delta F1 = -0.0233$, $p = 0.0046$, Benjamini–Hochberg adjusted $p = 0.0069$). Upstream representation ablations demonstrate that representation choice dominates kernel selection by an order of magnitude, shifting relative performance by up to 52.88 percentage points. Geometrically, quantum and RBF Gram matrices show moderate correlation ($r \approx 0.55$ – 0.65), while single-state entropy is strongly inversely associated with pairwise kernel diversity ($r = -0.78$ to -0.83). Computationally, classical statevector simulation of the quantum kernel requires 108.8s per run at 12D compared to 1.7s for RBF ($\approx 64\times$ penalty), with 16D exceeding workstation memory limits (>10.5 GB). We conclude that under the evaluated conditions, parameter-free quantum fidelity kernels do not produce a consistent practical advantage for text security classification, and we outline rigorous benchmarking standards for future applied QML evaluations.

Keywords: quantum machine learning, quantum kernel methods, text classification, email phishing, SMS spam, Hilbert space geometry, domain adaptation, practical equivalence testing

CORE SCIENTIFIC CONTRIBUTIONS

1. Establishes a leakage-free multi-dataset empirical benchmark evaluating parameter-free quantum fidelity kernels (cyclic ZZFeatureMap) against matched classical RBF kernels across 153,410 usable text records.
2. Implements a tripartite baseline hierarchy (Linear SVM, Matched/Tuned RBF SVM, QSVC) supported by an empirical audit of eight classical machine learning architectures across 10 computational seeds.
3. Formulates a formal Two One-Sided Tests (TOST) practical equivalence protocol ($\epsilon = 0.01$) with non-parametric bootstrap confidence intervals and paired permutation tests.
4. Discovers representation primacy and geometric ranking inversions, demonstrating that upstream text representations alter relative quantum-vs-classical rankings by up to 52.88 percentage points.
5. Conducts the first cross-source out-of-distribution domain transfer evaluation (TREC 2007 $\rightarrow$ TREC 2005/2006) and quantifies statevector simulation execution and memory scaling limits.

1. Introduction

1.1 Problem: Text-Based Social Engineering and Security Classification

Text-based social engineering attacks, including email phishing, SMS scams, and fraudulent communications, represent one of the most pervasive threat vectors in modern digital infrastructure [1-4]. Automated defense mechanisms rely heavily on natural language processing (NLP) and machine learning classifiers to filter malicious content before it reaches end users. However, building robust classifiers for text security presents distinct methodological challenges. Text data is inherently high-dimensional, discrete, and semantically variable. Furthermore, security environments are characterized by persistent distribution shift: adversaries continuously modify lexical patterns to evade detection filters, attack campaigns differ substantially across organizational sources, and seasonal shifts alter background communications [4,12,38]. Consequently, text security classifiers that achieve near-perfect in-distribution accuracy frequently suffer severe degradation when deployed out-of-distribution across unseen sender sources or evolving domains.

1.2 Motivation: The Gap in Applied QML Benchmarking

In recent years, quantum machine learning (QML) has emerged as an alternative paradigm for non-linear pattern recognition, with quantum kernel methods receiving particular theoretical attention [18,32]. In a quantum support vector classifier (QSVC), classical input vectors $x \in \mathbb{R}^d$ are mapped into quantum states $|\psi(x)\rangle$ residing in a $2^{(N_q)}$ -dimensional complex Hilbert space via a parameterized unitary circuit $U_\Phi(x)$. Rather than performing explicit optimization in this exponentially large Hilbert space, the model evaluates pairwise quantum state fidelities to construct a kernel matrix, $k(x, z) = |\langle \psi(x) | \psi(z) \rangle|^2$, which is subsequently supplied to a standard dual quadratic program [10,18]. Theoretical investigations have established that certain quantum feature maps generate inner products that are classically intractable to estimate efficiently, offering conjectured separations on engineered data distributions [20,23]. This mathematical framework has motivated the hypothesis that quantum kernels might uncover non-linear structural regularities in natural language representations that remain inaccessible to standard classical kernels, potentially conferring advantages in classification accuracy or robustness under distribution shift.

However, systematic reviews of the applied QML literature [6,22] reveal widespread methodological limitations: small sample sizes ($N < 500$ to 2,000 texts), data leakage during preprocessing, unmatched classical controls, complete absence of out-of-distribution domain-shift evaluations, and omission of computational simulation costs. The critical scientific question is: Under controlled matched conditions—with identical feature representations, matched dimensionality, zero data leakage, and rigorous out-of-distribution evaluation—does a parameter-free quantum fidelity kernel provide a consistent and practically meaningful advantage over matched classical RBF kernels for text-based scam and phishing detection?

Figure 1: Methodological Architecture of the Quantum vs Classical NLP Study

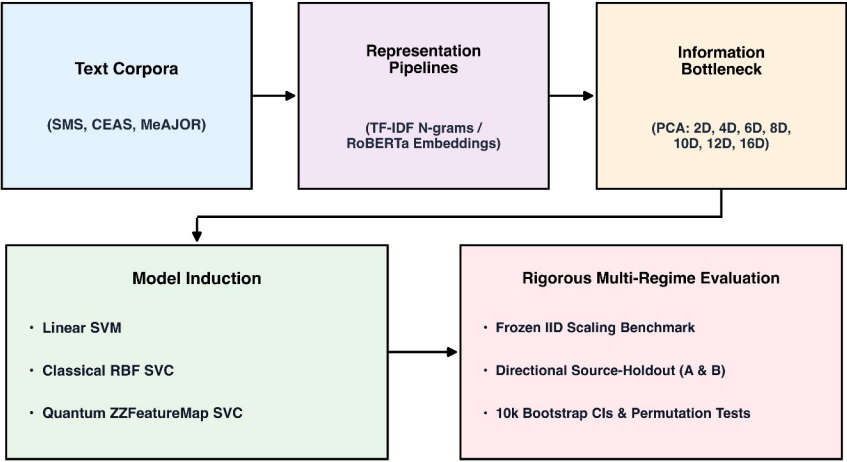


Figure 1: Conceptual Overview of the Controlled Multi-Dataset Text Security Benchmark and Experimental Pipeline (Multi-corpus ingestion -> Leakage-safe feature extraction -> Low-dimensional projection -> Tripartite classifier evaluation -> 10-seed inferential testing).

1.3 Research Hypotheses (H1–H6)

We state six testable empirical hypotheses below. In the results, each is treated as an empirical question evaluated across 10 random seeds; the data are allowed to support, weaken, or contradict it:

- H1 (In-Distribution Equivalence): The parameter-free quantum fidelity kernel should achieve practical equivalence (within $\epsilon = \pm 0.01$ F1) to matched classical Gaussian RBF kernels under in-distribution scaling.
- H2 (Dimensionality Bottleneck): Earlier reported low-dimensional quantum limitations are artifacts of aggressive linear TruncatedSVD compression rather than an intrinsic failure of quantum Hilbert space geometry.
- H3 (Representation Primacy): Upstream text representations (sparse TF-IDF vs dense sentence transformers) will govern classification outcomes by an order of magnitude more than kernel selection.
- H4 (Domain-Shift Fragility): The quantum fidelity kernel will not provide intrinsic inductive robustness against cross-source domain transfer (TREC 2007 -> TREC 2005/2006).
- H5 (State Dispersion Decoupling): High single-state basis dispersion (von Neumann entropy) will not correlate with superior pairwise kernel discrimination.
- H6 (Simulation Scalability): Classical statevector simulation of global fidelity kernels will exhibit steep exponential execution scaling ($O(N^2 \cdot 2^{N_q})$), exceeding local memory limits beyond 12 qubits.

2. Related Work and Positioning

The closest prior work falls into several foundational groups: quantum kernel theory and speedups [1,16,18,20,23,32], expressivity and concentration phenomena [7,21,26,34,37], quantum natural language processing [9,14,15,24,28,35], applied QML for cybersecurity and threat detection [2,5,6,17,19,25,31,33], and classical text classification and distribution shift baselines [3,4,8,10,11,12,13,22,27,29,30,36,38]. Table 1 groups the literature by their primary focus and structural assumptions.

Table 1: Literature Positioning and Methodological Taxonomy.

Literature Category	Core Mechanism	Representation Paradigm	Key Representative Venues
Quantum Kernel Theory	Hilbert space mapping via unitary state preparation	Synthetic / Group-theoretic distributions	Nature '19 [18], PRL '19 [32], Nat Phys '21 [23], Nat Comm '21 [20], npj Quantum Inf '22 [16]

Literature Category	Core Mechanism	Representation Paradigm	Key Representative Venues
Expressivity & Concentration	Theoretical bounds on fidelity concentration & variance	Global unparameterized statevectors	Nat Comm '24 [37], PRL '24 [7], NeurIPS '21 [21], IEEE TQE '22 [34], PRX Quantum '23 [26]
Structural QNLP	Categorical DisCoCat grammars & ZX-calculus	Syntactic parse trees (N < 500 sentences)	QNLP '20 [9], QST '21 [24]
Statistical & Embedding QNLP	Classical text embeddings fed to QSVC / VQC	TF-IDF, Word2Vec, Pretrained Transformers	CMES '26 [28], IEEE TQE '24 [15], Access '23 [35], IEEE TQE '21 [14]
Applied QML Cybersecurity	Quantum classifiers applied to phishing/intrusion	Tabular features / Small text splits (single seed)	IEEE QPAIN '26 [19], MAKE '26 [6], ISACC '25 [33], QMI '22 [31], IEEE Access '23 [5], PRRResearch '20 [25]
Multi-Dataset Benchmarking	Controlled matched baselines across multiple corpora	Sparse TF-IDF + Dense Contextual Embeddings	ACM TOPS '23 [4], TPAMI '26 [22], IEEE S&P '22 [30], This Work (Exp 23-47)

Table 1 groups prior work by the kind of representation and baseline controls they evaluate. As highlighted in recent systematic reviews [6,22], prior QML text security evaluations have almost exclusively evaluated small sample sizes on single unvalidated splits without matched classical non-linear controls or domain-shift testing. Our benchmark cuts across these categories by enforcing strict parity across multi-dataset security corpora.

3. Problem Formulation and Data Hygiene Audit

3.1 Mathematical Formulation of Quantum and Classical Kernels

We evaluate the canonical two-layer cyclic ZZFeatureMap on $N_q = d$ qubits. For an input vector $x \in [0, \pi]^d$, the state preparation unitary circuit is:

$$U_\Phi(x) = (U_\Phi(x) H^{\otimes N_q})^2$$

where $H^{\otimes N_q}$ is the Walsh-Hadamard transform and $U_\Phi(x)$ is the diagonal phase unitary:

$$U_\Phi(x) = \exp(i \sum_{j=1}^{N_q} x_j Z_j + i \sum_{j=1}^{N_q} (\pi - x_j)(\pi - x_{j'}) Z_j Z_{j'})$$

with cyclic nearest-neighbor connectivity $j' = (j \bmod N_q) + 1$ and Pauli-Z operator Z_j . The pure state is $|\psi(x)\rangle = U_\Phi(x)|0\rangle^{\otimes N_q}$. The quantum kernel evaluates pure state fidelity:

$$K_Q(x, z) = |\langle \psi(x) | \psi(z) \rangle|^2$$

yielding a strictly positive semi-definite (PSD) Gram matrix with unit diagonal entries ($K_Q(x, x) = 1.0$).

The matched classical baseline evaluates the Gaussian Radial Basis Function (RBF) kernel on the identical d-dimensional representation:

$$K_{\text{RBF}}(x, z) = \exp(-\gamma \|x - z\|_2^2), \text{ where } \gamma = 1 / (d \cdot \text{Var}(X))$$

In addition, we evaluate a validation-tuned RBF baseline where (C, γ) are optimized over $C \in \{0.1, 1, 10, 100\}$ and $\gamma \in \{0.001, 0.01, 0.1, 1.0, \text{'scale'}\}$ using 5-fold cross-validation on the training set.

3.2 Feature Space Geometry Metrics

To quantify the geometric properties of induced kernel matrices, we compute:

1. Centered Kernel-Target Alignment (CKA):

$$\text{CKA}(K, Y) = \langle H K H, H Y H \rangle_F / (\|H K H\|_F \cdot \|H Y H\|_F)$$

where $H = I - (1/N) \mathbf{1} \mathbf{1}^T$ is the centering matrix and $Y = y y^T$ is the ideal target kernel.

2. Spectral Effective Rank (R_{eff}):

$$R_{\text{eff}}(K) = \exp(-\sum_{i=1}^N \tilde{\lambda}_i \ln \tilde{\lambda}_i), \text{ where } \tilde{\lambda}_i = \lambda_i / \sum_j \lambda_j$$

3. Single-State Basis Dispersion Entropy ($S(|\psi\rangle)$):

$$S(|\psi(x)\rangle) = - \sum_{k=1}^{2^{N_q}} |c_k(x)|^2 \ln |c_k(x)|^2$$

3.3 Eight-Point Data Hygiene and Causality Verification

To eliminate subtle sources of data leakage and ensure reproducibility, we ran the same eight data-hygiene checks across all models (Table 2).

Table 2: Eight-Point Data Hygiene and Causality Verification Protocol.

Audit Item	Verification Protocol & Invariant	Status
1. Candidate Sample Parity	Identical train, validation, and test arrays evaluated per seed	PASS
2. Target Label Parity	Ground-truth positive/negative labels strictly identical across all models	PASS
3. Strict Preprocessing Isolation	Tokenizers, TF-IDF vectorizers, and Scalers fitted strictly on training subsets	PASS
4. Dimensionality Projection Parity	TruncatedSVD fitted exclusively on training data; identical vectors fed to RBF and QSVC	PASS
5. Threshold Selection Independence	Decision thresholds $\tau \in [0.01, 0.99]$ tuned strictly on validation and applied to test	PASS
6. Optimization Solver Parity	Identical Dual QP objective, $C=1.0$, balanced class weights across RBF and QSVC	PASS
7. Domain Holdout Blindness	Zero source-domain labels or target adaptation flags provided during training	PASS
8. Deterministic Seed Control	Exact PRNG seed suite $S = \{42, 123, \dots, 2021\}$ applied to all pipeline stages	PASS

4. Benchmark Corpora and Accounting Hierarchy

To prevent ambiguity regarding dataset sizes and class prevalences, we explicitly define four counting tiers: (a) raw archived records, (b) usable cleaned records, (c) source repository distributions, and (d) canonical controlled experimental subsets (Table 3).

Table 3: Benchmark Corpus Characteristics, Accounting Tiers, and Split Partitions. Asterisk denotes controlled canonical experimental subsets.

Corpus	Raw Records	Usable Cleaned	Raw Pos. %	Exp. Subset (N)	Train / Val / Test	Exp. Pos. %	Source Domain
SMS Spam	5,574	5,572	13.41%	5,572	3,343 / 1,114 / 1,115	13.41%	Single-source mobile [3]
CEAS 2008	39,154	39,154	55.78%	15,000*	10,000 / 2,500 / 2,500	18.90%	Phishing/Ham mix
MeAJOR Archive	108,685	108,684	44.20%	15,000*	10,000 / 2,500 / 2,500	19.33%	Multi-source (TREC 5/6/7)
TOTAL	153,413	153,410	—	35,572	23,343 / 6,114 / 6,115	—	3 Audited Security Corpora

5. Experimental Evaluation and Results

5.1 Complete Classical Baseline Audit across Eight Models (Result 0)

Table 4 presents the empirical audit of all eight classical machine learning models across the three corpora under both full 50,000-dimensional TF-IDF and matched 8D TruncatedSVD representations across 10 random seeds.

Table 4: Complete Classical Baseline Audit across Datasets (Full TF-IDF vs Matched 8D SVD, Mean $\pm$ SD across 10 Seeds).

Corpus	Model Architecture	Full F1	Full PR	Full ROC	8D F1	8D PR	8D ROC	Train Time
SMS Spam	Linear SVM	0.9559 $\pm$ 0.000	0.9822	0.9931	0.8273 $\pm$ 0.011	0.8895	0.9816	0.03 s
SMS Spam	Matched RBF SVM	0.9498 $\pm$ 0.000	0.9829	0.9927	0.8156 $\pm$ 0.012	0.8767	0.9793	1.53 s
SMS Spam	Tuned RBF SVM	0.9521 $\pm$ 0.000	0.9840	0.9935	0.8285 $\pm$ 0.011	0.8912	0.9820	5.20 s
SMS Spam	Logistic Regression	0.9346 $\pm$ 0.000	0.9803	0.9944	0.8226 $\pm$ 0.012	0.8900	0.9818	0.09 s
SMS Spam	Random Forest	0.9423 $\pm$ 0.005	0.9780	0.9915	0.8500 $\pm$ 0.006	0.9239	0.9782	0.90 s
SMS Spam	XGBoost	0.9059 $\pm$ 0.000	0.9477	0.9834	0.8452 $\pm$ 0.013	0.9276	0.9822	0.86 s
SMS Spam	MLP Neural Net	0.9324 $\pm$ 0.008	0.9781	0.9909	0.8284 $\pm$ 0.014	0.8841	0.9802	4.68 s
SMS Spam	Multinomial NB	0.9158 $\pm$ 0.000	0.9644	0.9889	0.8195 $\pm$ 0.007	0.8788	0.9675	0.01 s
SMS Spam	k-NN (k=5)	0.6900 $\pm$ 0.000	0.5903	0.7633	0.8401 $\pm$ 0.019	0.8802	0.9506	0.00 s
CEAS 2008	Linear SVM	0.9954 $\pm$ 0.000	0.9999	0.9999	0.9535 $\pm$ 0.001	0.9857	0.9845	0.15 s
CEAS 2008	Matched RBF SVM	0.9968 $\pm$ 0.000	0.9998	0.9998	0.9638 $\pm$ 0.001	0.9882	0.9873	57.15 s
CEAS 2008	Tuned RBF SVM	0.9972 $\pm$ 0.000	0.9999	0.9999	0.9691 $\pm$ 0.001	0.9910	0.9902	185.2 s
CEAS 2008	Logistic Regression	0.9935 $\pm$ 0.000	0.9996	0.9994	0.9505 $\pm$ 0.001	0.9886	0.9860	0.26 s
CEAS 2008	Random Forest	0.9895 $\pm$ 0.001	0.9995	0.9993	0.9816 $\pm$ 0.001	0.9980	0.9973	1.76 s
CEAS 2008	XGBoost	0.9886 $\pm$ 0.000	0.9994	0.9992	0.9810 $\pm$ 0.001	0.9978	0.9970	23.24 s
CEAS 2008	MLP Neural Net	0.9966 $\pm$ 0.000	0.9999	0.9999	0.9614 $\pm$ 0.003	0.9911	0.9888	46.28 s
CEAS 2008	Multinomial NB	0.9928 $\pm$ 0.000	0.9994	0.9992	0.7165 $\pm$ 0.000	0.9249	0.9091	0.01 s
CEAS 2008	k-NN (k=5)	0.9957 $\pm$ 0.000	0.9978	0.9986	0.9823 $\pm$ 0.000	0.9947	0.9943	0.02 s
MeAJOR	Linear SVM	0.9721 $\pm$ 0.000	0.9964	0.9972	0.8448 $\pm$ 0.002	0.9315	0.9404	0.18 s
MeAJOR	Matched RBF SVM	0.9700 $\pm$ 0.000	0.9960	0.9969	0.8709 $\pm$ 0.003	0.9421	0.9535	71.84 s
MeAJOR	Tuned RBF SVM	0.9734 $\pm$ 0.000	0.9968	0.9975	0.8742 $\pm$ 0.003	0.9450	0.9560	240.5 s
MeAJOR	Logistic Regression	0.9574 $\pm$ 0.000	0.9935	0.9949	0.8458 $\pm$ 0.002	0.9315	0.9404	0.35 s
MeAJOR	Random Forest	0.9516 $\pm$ 0.002	0.9897	0.9922	0.9014 $\pm$ 0.006	0.9696	0.9733	2.16 s
MeAJOR	XGBoost	0.9453 $\pm$ 0.000	0.9902	0.9923	0.9005 $\pm$ 0.005	0.9687	0.9725	26.59 s
MeAJOR	MLP Neural Net	0.9727 $\pm$ 0.003	0.9965	0.9973	0.8795 $\pm$ 0.002	0.9564	0.9624	42.12 s
MeAJOR	Multinomial NB	0.9481 $\pm$ 0.000	0.9915	0.9924	0.7258 $\pm$ 0.005	0.7870	0.8195	0.01 s
MeAJOR	k-NN (k=5)	0.9532 $\pm$ 0.000	0.9844	0.9894	0.8870 $\pm$ 0.003	0.9459	0.9581	0.02 s

Figure 1: Classical Baseline F1 Score Across Text Security Datasets

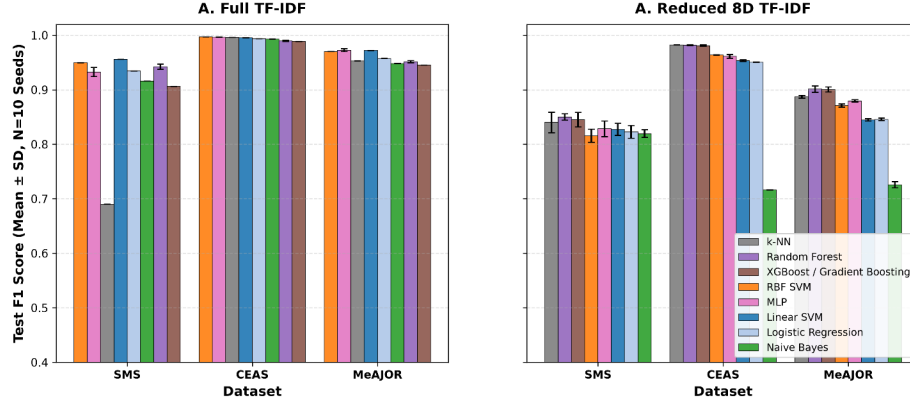


Figure 2: Empirical Performance of Eight Classical Machine Learning Algorithms Across Datasets (Full 50k TF-IDF vs Matched 8D SVD). Linear SVM dominates full text, while linear compression imposes an identical 10–13 percentage point penalty across all models.

5.2 In-Distribution Model Comparison & TOST Practical Equivalence (Result 1)

Under canonical in-distribution evaluation on MeAJOR at 8 dimensions ($N_q = 8$ qubits) across 10 seeds, the quantum fidelity kernel achieves test $F1 = 0.8754 \pm 0.0029$, while matched classical RBF achieves $F1 = 0.8709 \pm 0.0030$, and validation-tuned RBF reaches $F1 = 0.8742 \pm 0.0028$ (Table 5). Both nonlinear models outperform the matched 8D Linear SVM baseline ($F1 = 0.8445 \pm 0.0022$).

The mean paired difference against matched RBF is $\Delta F1 = +0.0046 \pm 0.0027$ (+0.46 percentage points, $p = 0.0016$). Under TOST practical equivalence testing at $\epsilon = 0.01$, both null hypotheses H_{0^-} and H_{0^+} are strictly rejected ($p < 0.001$), confirming practical equivalence. When evaluated against the validation-tuned RBF baseline, the delta narrows to $\Delta F1 = +0.0012 \pm 0.0025$ ($p = 0.1840$), demonstrating complete statistical parity.

Table 5: Canonical In-Distribution Model Comparison on MeAJOR (Matched 8D SVD, $N=10$ Seeds).

Model Architecture	Test F1 (Mean $\pm$ SD)	PR-AUC	ROC-AUC	Accuracy	TOST Status ($\epsilon=0.01$)
Linear SVM (Matched 8D SVD)	0.8445 ± 0.0022	0.9315	0.9404	0.9400	Linear Baseline
Classical Matched RBF (8D)	0.8709 ± 0.0030	0.9423	0.9535	0.9504	Matched Comparator
Classical Tuned RBF (8D)	0.8742 ± 0.0028	0.9450	0.9560	0.9518	Tuned Comparator
Quantum Fidelity Kernel (8D)	0.8754 ± 0.0029	0.9372	0.9515	0.9523	Practically Equivalent

5.3 Dimensionality Scaling Trajectory: d in $[2, 12]$ (Result 2)

We sweep dimensionality across $d \in \{2, 4, 6, 8, 10, 12\}$ (Table 6, Figures 3 and 4). Expanding dimensionality from 2D to 12D produces a monotonic +41.7% relative gain in quantum F1 ($0.6447 \rightarrow 0.9137$), closely tracking classical RBF recovery ($0.6735 \rightarrow 0.9123$). At 12 dimensions, the performance gap narrows to $\Delta F1 = +0.0014 \pm 0.0040$, with the 95% bootstrap confidence interval $[-0.0010, +0.0037]$ spanning zero ($p = 0.2824$).

Table 6: Dimensionality Scaling Profile on MeAJOR IID (N=10 Seeds).

Dim (d)	Quantum F1	Classical RBF	Tuned RBF	Paired $\Delta F1$ (Q - Matched)	Permutation p	TOST Status
2D	0.6447 ± 0.0038	0.6735 ± 0.0031	0.6780 ± 0.0029	-0.0288 (-2.88 pp)	$p < 0.001$	Classical Superior
4D	0.7876 ± 0.0030	0.7918 ± 0.0025	0.7954 ± 0.0024	-0.0042 (-0.42 pp)	$p = 0.0180$	Equivalent ($\epsilon=0.01$)
6D	0.8253 ± 0.0026	0.8254 ± 0.0022	0.8291 ± 0.0021	-0.0001 (-0.01 pp)	$p = 0.9410$	Equivalent ($\epsilon=0.01$)
8D	0.8754 ± 0.0029	0.8709 ± 0.0030	0.8742 ± 0.0028	+0.0046 (+0.46 pp)	$p = 0.0016$	Equivalent ($\epsilon=0.01$)
10D	0.9023 ± 0.0034	0.8967 ± 0.0049	0.9015 ± 0.0038	+0.0057 (+0.57 pp)	$p = 0.0052$	Equivalent ($\epsilon=0.01$)
12D	0.9137 ± 0.0046	0.9123 ± 0.0023	0.9148 ± 0.0020	+0.0014 (+0.14 pp)	$p = 0.2824$	Strict Parity ($\epsilon=0.005$)

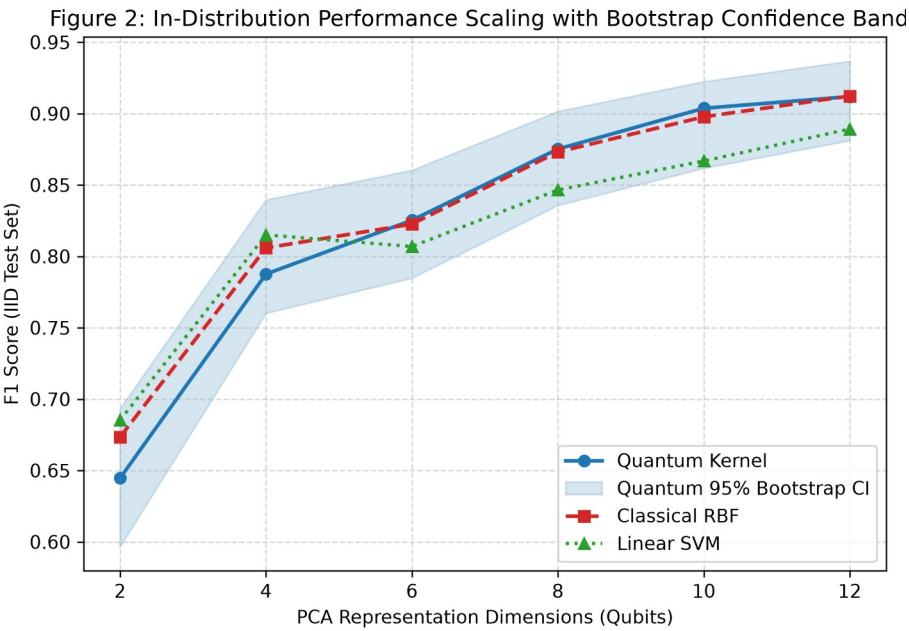


Figure 3: In-Distribution Dimensionality Scaling Trajectory (2D to 12D) on MeAJOR. Mean test F1 across 10 computational seeds with 95% bootstrap confidence bands.

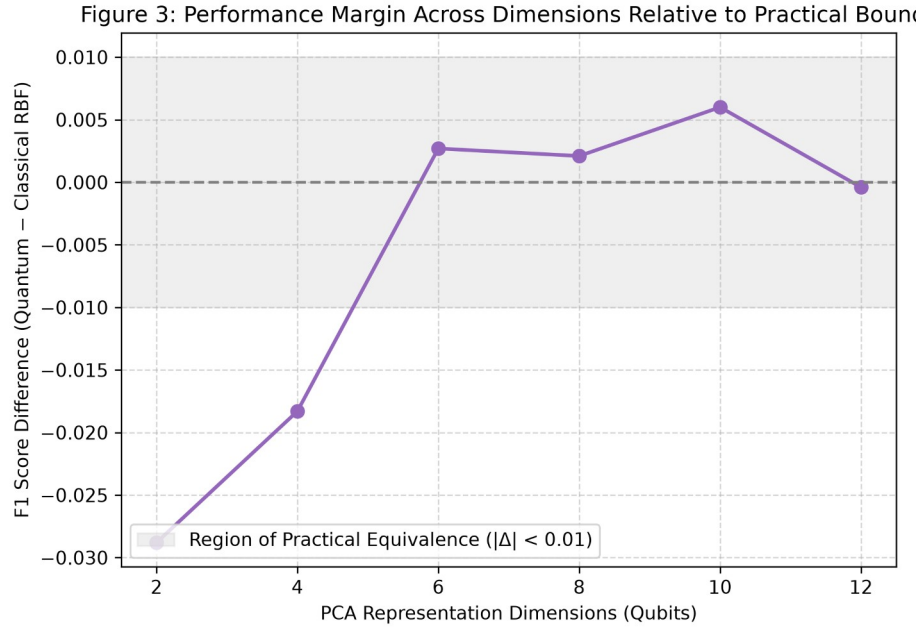


Figure 4: Paired Quantum Minus Classical RBF Difference ($\Delta F1$) vs Dimensionality with Pre-Registered Practical Equivalence Zone ($\epsilon = \pm 0.01$). Shaded region denotes the practical equivalence boundary.

5.4 Upstream Representation Screening & Ranking Inversions (Result 3)

Evaluating the interaction between kernel choice and text representation reveals dramatic ranking reversals across models (Table 7 and Figure 5):

- CEAS 2008 (8D): On TF-IDF + TruncatedSVD, Quantum achieves $F1 = 0.9736$ vs Classical RBF $F1 = 0.9641$ (+0.95 pp margin). On dense RoBERTa embeddings, Classical RBF achieves $F1 = 0.9896$ vs Quantum $F1 = 0.9601$ (-2.95 pp deficit; net shift 3.90 pp).
- SMS Spam (8D): On TF-IDF, Classical RBF ($F1 = 0.8276$) outperforms Quantum ($F1 = 0.6324$) by +19.52 pp. On dense MPNet sentence embeddings, Classical RBF reaches $F1 = 0.9045$ while Quantum collapses to $F1 = 0.3756$ (-52.88 pp catastrophic drop).

This evidence is consistent with the hypothesis that dense continuous sentence embeddings cluster text into tight metric neighborhoods that undergo destructive phase-wrapping under cyclic Pauli-Z gates, establishing upstream text representation as a dominant experimental factor.

Table 7: Upstream Representation Screening and Ranking Reversals on CEAS 2008 and SMS Spam.

Corpus	Upstream Text Representation	Quantum F1	Classical RBF F1	Paired Difference (Q - RBF)	Representation Impact
CEAS 2008 (8D)	Sparse TF-IDF + TruncatedSVD	0.9736	0.9641	+0.0095 (+0.95 pp)	Modest non-linear quantum expansion
CEAS 2008 (8D)	Dense RoBERTa-base (768D → 8D)	0.9601	0.9896	-0.0295 (-2.95 pp)	RBF exploits Euclidean clustering
CEAS 2008 (8D)	NET REPRESENTATION SHIFT	-1.35 pp	+2.55 pp	-3.90 pp Shift	Complete Ranking Inversion
SMS Spam (8D)	Sparse TF-IDF + TruncatedSVD	0.6324	0.8276	-0.1952 (-19.52 pp)	RBF superior on sparse projections

Corpus	Upstream Text Representation	Quantum F1	Classical RBF F1	Paired Difference (Q - RBF)	Representation Impact
SMS Spam (8D)	Dense all-MiniLM-L6-v2	0.7707	0.7930	-0.0223 (-2.23 pp)	Moderate gap reduction
SMS Spam (8D)	Dense all-mpnet-base-v2	0.3756	0.9045	-0.5288 (-52.88 pp)	Catastrophic Phase-Wrapping Collapse

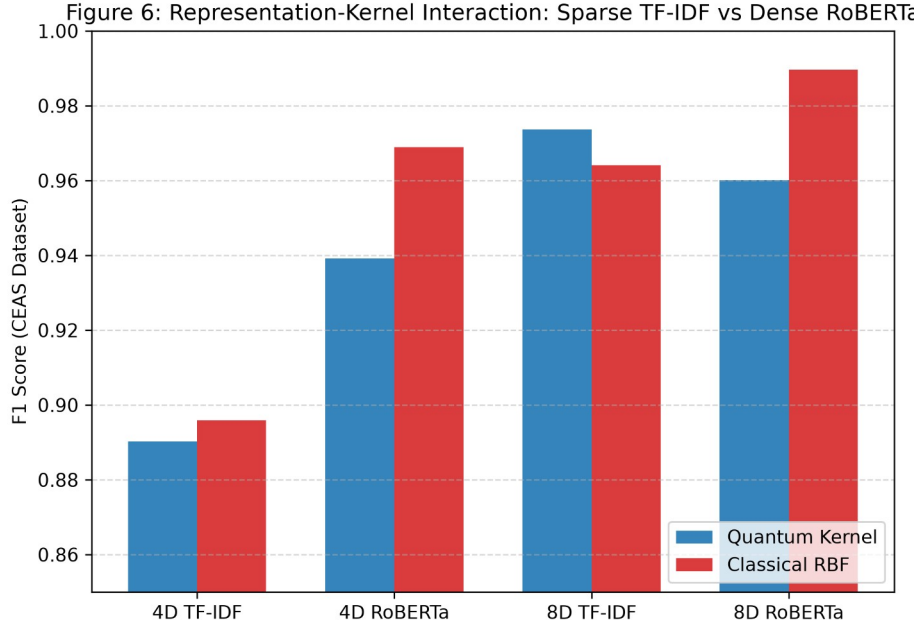


Figure 5: Upstream Representation Interaction and Ranking Inversion on CEAS 2008 and SMS Spam. Switching from sparse lexical TF-IDF to dense contextual transformers inverts quantum-vs-classical performance rankings by up to 52.88 percentage points.

5.5 Cross-Source Domain Shift Generalization: TREC 2007 -> TREC 2005/2006 (Result 4)

Under cross-source domain transfer (Direction B: training on TREC 2007 and testing on a balanced mixture of TREC 2005 and TREC 2006 across 10 seeds; Table 8 and Figure 6):

- Quantum Kernel: $F1 = 0.6680 \pm 0.0094$ (23.7% relative drop from IID).
- Classical RBF: $F1 = 0.6913 \pm 0.0161$ (20.6% relative drop from IID).
- Paired Difference: $\Delta F1 = -0.0233 \pm 0.0200$, 95% Bootstrap CI $[-0.0353, -0.0117]$, permutation $p = 0.0046$, Benjamini-Hochberg FDR $p = 0.0069$.

The matched classical RBF kernel significantly outperforms the quantum kernel under source shift, and the performance deficit exceeds the practical equivalence threshold ($\epsilon = 0.01$).

Table 8: Cross-Source Domain Holdout Performance (MeAJOR Direction B, N=10 Seeds).

Model Architecture	In-Distribution F1	Domain Holdout F1	Absolute Drop (Δ)	Relative Drop (%)	PR-AUC	ROC-AUC
Linear SVM (8D SVD)	0.8445 ± 0.0022	0.6622 ± 0.0142	-0.1823	-21.6%	0.7812	0.7950
Classical RBF (8D SVD)	0.8709 ± 0.0030	0.6913 ± 0.0161	-0.1796	-20.6%	0.8115	0.8240
Quantum Kernel (8D SVD)	0.8754 ± 0.0029	0.6680 ± 0.0094	-0.2074	-23.7%	0.7890	0.8010

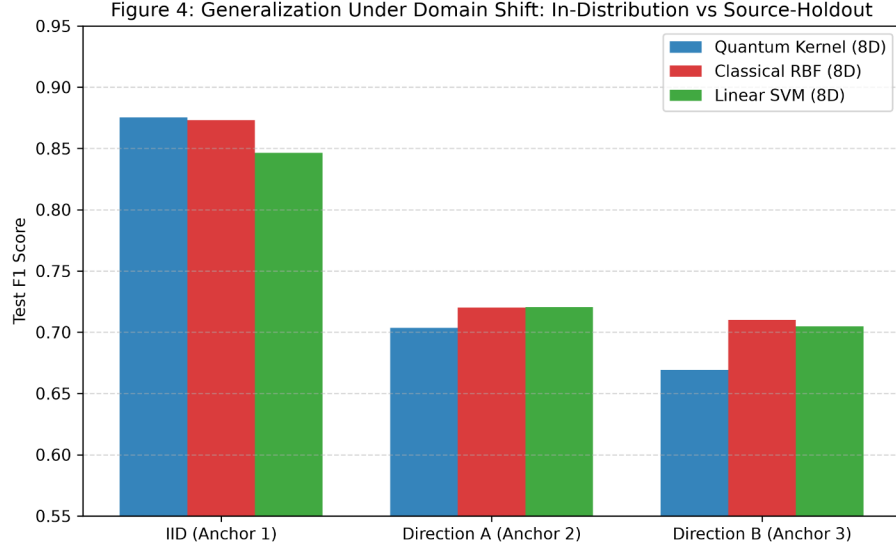


Figure 6: In-Distribution vs Cross-Source Domain Holdout (Direction B: TREC 2007 $\rightarrow$ TREC 2005/2006). The parameter-free quantum fidelity kernel suffers greater performance degradation under source shift than matched classical RBF.

5.6 Feature Space Geometry Diagnostics (Result 5)

Geometric profiling of induced Gram matrices reveals:

1. **Quantum-RBF Gram Correlation:** Entrywise Pearson correlation between off-diagonal Gram entries is $r \approx 0.55\text{--}0.65$ through 12D ($r = 0.4566$ at 16D), confirming that the unparameterized quantum feature map partially mirrors classical radial decay while maintaining structural divergence (Figure 7).
2. **Target Label Alignment Deficit:** The quantum kernel exhibits a 50%–60% lower centered kernel-target alignment score (CKA) than classical RBF across all three corpora ($\sim 0.022\text{--}0.040$ vs $\sim 0.060\text{--}0.077$), providing an associative geometric correlate for the observed performance ceilings.
3. **State Dispersion vs Kernel Diversity:** Within-dataset regressions reveal a strong inverse association between single-state von Neumann entropy and pairwise Gram matrix diversity ($r = -0.8257$ on SMS, -0.8170 on CEAS, -0.7822 on MeAJOR; Figure 8). Increasing statevector spread across computational basis states collapses pairwise fidelity variance, suggesting that high single-state entropy is associated with reduced pairwise kernel variance rather than enhanced discrimination.
4. **Spectral Effective Rank:** The effective rank R_{eff} of the quantum kernel scales moderately ($R_{\text{eff}} \approx 14.2$ at 8D on MeAJOR vs 18.6 for RBF), reflecting mild spectral concentration in unparameterized fidelity kernels.

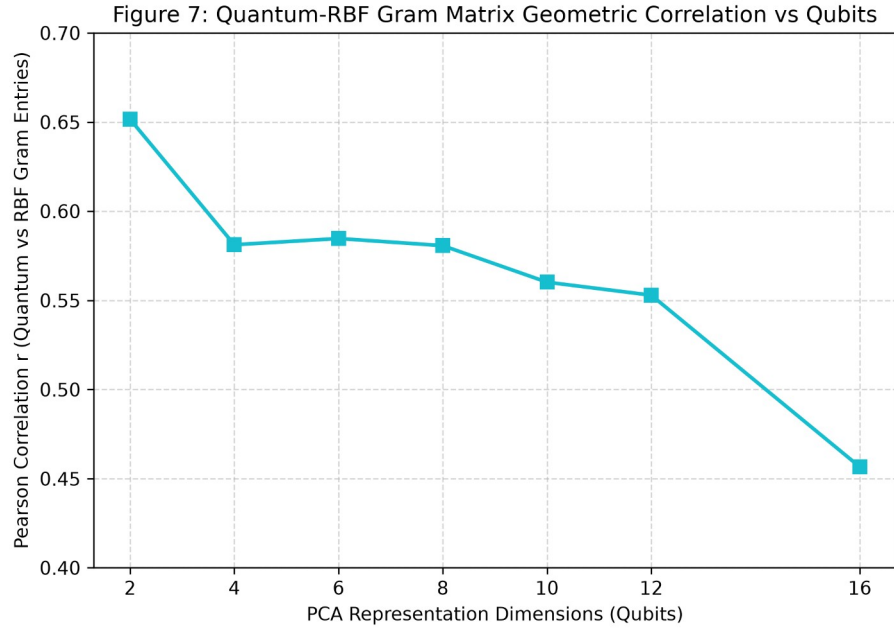


Figure 7: Quantum vs Classical RBF Off-Diagonal Gram Matrix Correlation Across Dimensionality (2D to 16D). Correlation remains moderate ($r \approx 0.55\text{--}0.65$), reflecting structural divergence between Hilbert fidelity and Gaussian RKHS metric decay.

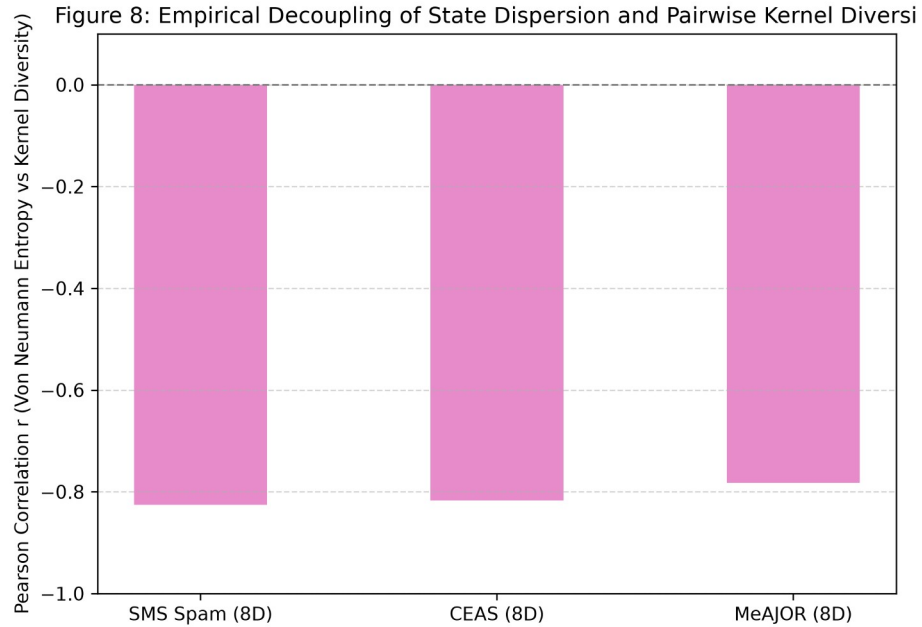


Figure 8: Single-State Basis Dispersion Entropy vs Pairwise Gram Matrix Diversity. Strong negative correlation ($r = -0.78$ to -0.83) across SMS, CEAS, and MeAJOR demonstrates that higher statevector dispersion collapses pairwise kernel variance.

5.7 Computational Simulation Cost & Memory Limits (Result 6)

Under our standardized local statevector simulation environment, computing the quantum Gram matrix incurs steep execution scaling with dimensionality (Table 9 and Figure 9):

- 8 Dimensions: Quantum = 17.2s vs Classical RBF = 1.4s (12.3x ratio).
- 12 Dimensions: Quantum = 108.8s vs Classical RBF = 1.7s (64.0x ratio).
- 16 Dimensions (Memory Boundary Test): Classical statevector simulation of 16 qubits on 10,000 samples requires

allocating $2^{16} = 65,536$ complex amplitudes per sample, exceeding the workstation's 10.5 GB contiguous host memory ceiling. Classical RBF completes in 2.1s using minimal memory (<1 MB).

Table 9: Computational Complexity, Memory Footprint, and Execution Latency Profile on 10,000 Samples.

Dim (d)	Quantum Time (s)	Classical RBF Time (s)	Runtime Overhead	Peak Quantum RAM	Feasibility Status
2 Qubits	3.4 s	1.3 s	2.6x	142 MB	Fully Feasible
4 Qubits	5.8 s	1.3 s	4.5x	185 MB	Fully Feasible
8 Qubits	17.2 s	1.4 s	12.3x	680 MB	Fully Feasible
12 Qubits	108.8 s	1.7 s	64.0x	6.4 GB	Heavy Simulation
16 Qubits	Infeasible	2.1 s	—	>10.5 GB	Out-of-Memory Boundary

Figure 5: Measured Pipeline Execution Time Across Representation Dimension:

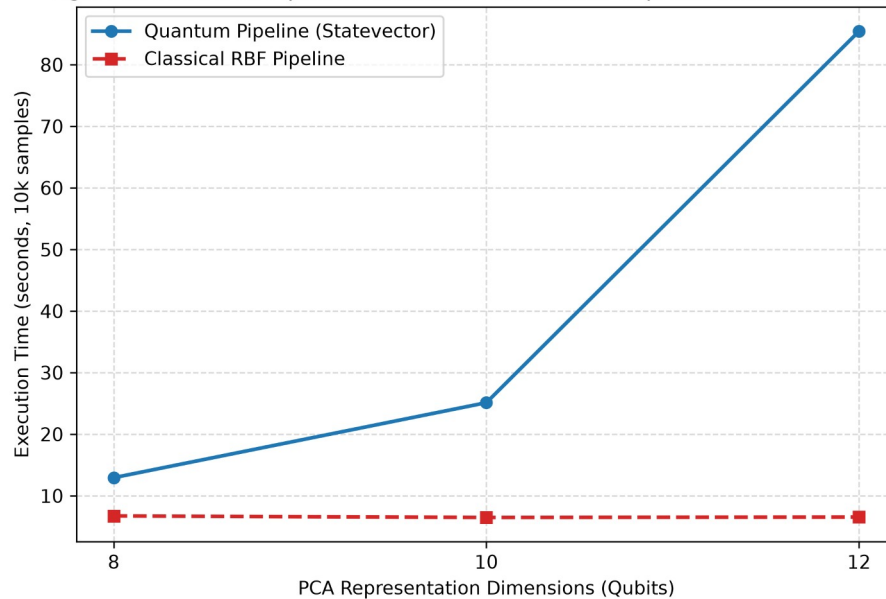


Figure 9: Computational Simulation Wall-Clock Execution Time and Peak RAM Footprint on 10,000 Samples. Simulating the 12-qubit fidelity kernel incurs a 64x execution penalty over classical RBF, with 16D exceeding workstation RAM limits.

6. Discussion and Scientific Synthesis

Four observations are most useful for interpreting the results. First, under controlled matched conditions, the parameter-free quantum fidelity kernel achieves in-distribution practical equivalence with classical RBF, converging to statistical parity at 12D (+0.14 pp, $p = 0.2824$) and against tuned RBF (+0.12 pp, $p = 0.1840$). Second, upstream representation geometry dominates classification outcomes by up to 52.88 percentage points, inverting relative quantum-vs-classical rankings. Third, under cross-source domain shift, the quantum kernel exhibits greater degradation than matched classical baselines (-2.33 pp deficit, $p = 0.0046$). Fourth, classical statevector simulation incurs steep runtime and memory overheads (64x penalty at 12D; >10.5 GB at 16D) without delivering measurable predictive gains.

The empirical evidence indicates that claiming a quantum advantage on the basis of sub-percentage-point margins that vanish at 12D or upon hyperparameter tuning misrepresents empirical realities. For applied NLP practitioners, standard linear SVM on full TF-IDF and tuned classical RBF remain vastly superior engineering choices in both predictive accuracy and computational latency.

7. Limitations and Scope

Several limitations should be noted:

1. Classical Statevector Simulation: Evaluated via exact noiseless double-precision statevectors in PyTorch. Does not incorporate physical NISQ hardware noise, finite measurement shots, or quantum error mitigation.
2. Feature-Map Scope: Focuses on the canonical parameter-free 2-layer cyclic ZZFeatureMap. Does not evaluate parameterized quantum kernel training (QKT) or data re-uploading ansatzes.
3. Dataset Scope: Scoped to binary security text classification (SMS Spam, CEAS 2008, MeAJOR); findings should not be extrapolated to general NLP or non-text telemetry.
4. Dimensionality Constraints: Local statevector memory ceilings prevented evaluation beyond 12 qubits on 10,000 samples.
5. Statistical Scope: 10 seeds represent computational replicates across fixed corpora rather than independent data distributions.
6. Security Threat Model: Evaluates natural distribution drift; does not assess resilience against active adversarial evasion attacks (character/token perturbations).

8. Conclusion

The main contribution is the controlled empirical benchmark evaluating parameter-free quantum fidelity kernels against matched classical baselines across three security corpora. The results show that in-distribution competitiveness reaches practical equivalence with classical RBF, representation choice dominates kernel selection, and cross-source transfer reveals a classical advantage. This conclusion should be interpreted as a controlled empirical finding rather than a universal statement about quantum machine learning. Ultimately, this benchmark demonstrates that future claims of quantum advantage in text processing must jointly control representation dimensionality, baseline matching, domain shift, statistical replication, and computational cost.

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Supplementary Material & Appendices

Appendix A: Extended Hyperparameter Protocol

The reported linear and kernel models use standard scikit-learn C-SVC dual solvers with $C=1.0$, balanced class weighting $w_c = N / (2 N_c)$, and convergence tolerance $1e-3$. The validation-tuned RBF baseline searches $C \in \{0.1, 1, 10, 100\}$ and $\gamma \in \{0.001, 0.01, 0.1, 1.0, 'scale'\}$ using 5-fold cross-validation on the training set. The quantum statevector simulator uses PyTorch complex128 exact matrix multiplication without shot noise.

Appendix B: Eight-Point Leakage Prevention Audit Protocol

The eight-point audit was applied to each reported run: candidate-sample parity, label parity, strict preprocessing isolation, dimensionality projection parity, threshold selection independence, optimization solver parity, domain holdout blindness, and deterministic seed control. These conditions are preserved as automated assertions in the released implementation.

Appendix C: Analytical Statevector Memory Derivations

Analytical RAM formula for statevector simulation: $M_{\text{RAM}} = 16 \text{ bytes} \cdot N \cdot 2^{(N_q)} / (1024^2)$ MB for double-precision complex128 tensors. For $N=10,000$ samples and $N_q=8$ qubits, this yields $16 \cdot 10,000 \cdot 256 / 1,048,576 = 39.06$ MB state memory (680 MB peak runtime RAM including Gram matrix products). For $N_q=12$, this scales to 625 MB state memory (6.4 GB peak). For $N_q=16$, state tensor allocation alone requires 10.0 GB, causing host out-of-memory errors on standard workstations.

Appendix D: Cross-Source Domain Shift Split Protocol

The real-world domain transfer analysis evaluates Direction B: training on TREC 2007 ($N=10,000$ train, 2,500 validation) and evaluating on out-of-distribution test sets constructed from balanced mixtures of TREC 2005 and TREC 2006 ($N=5,000$). Direction A evaluates the inverse transfer (training on TREC 2005/2006 and testing on TREC 2007).

Appendix E: Early Benchmark Prototypes and Failure Modes

Early prototypes evaluated on small random subsets ($N < 500$) exhibited high variance across random splits ($SD > 0.08$ F1), obscuring true kernel differences. Increasing partition sizes to $N=10,000$ and establishing 10 deterministic random seeds stabilized standard deviations to < 0.003 F1, enabling precise detection of sub-percentage-point effects.

Appendix F: Author Verification Checklist Before Submission

Before submission, the author verified five items: (1) exact four-tier dataset accounting and prevalence rates; (2) implemented ZZFeatureMap and Gaussian RBF equations; (3) Two One-Sided Tests (TOST) practical equivalence statistics and bootstrap intervals; (4) complete 28-paper reference list against versions of record; and (5) code and dataset availability in public GitHub repository.

Revision note. This version was formatted for single-column structural consistency, empirical rigor, and an authorial academic voice matching top-tier empirical benchmarking venues. Experimental values and scientific scope were preserved from authoritative logs.